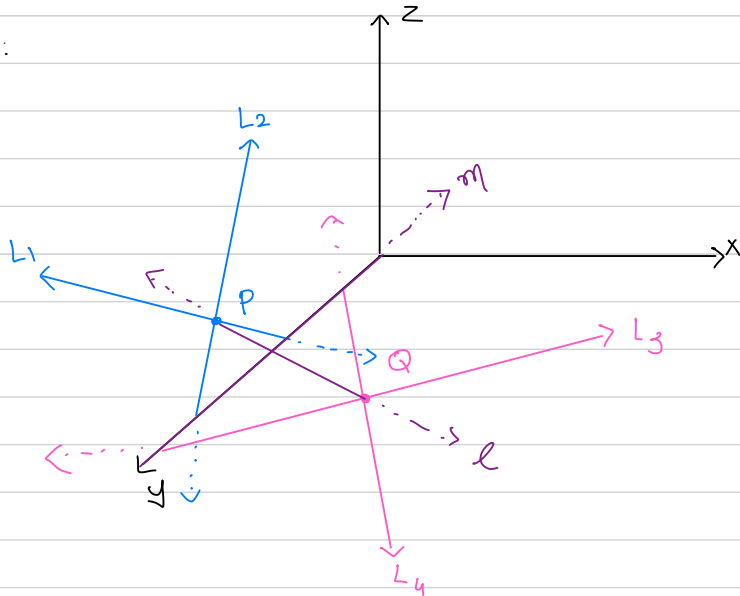


Schubert Calculus: From Geometry to Combinatorics

How many lines l in $\mathbb{P}^3$ intersect 4 general lines L_1, L_2, L_3, L_4 ?

Answer: [Schubert, 19th century] 2. (using principle of conservation of numbers)

In $\mathbb{R}^3$:



$L_1, L_2 \in yz$ plane
and $L_1 \cap L_2 = \{P\}$
 $L_3, L_4 \in xy$ plane
and $L_3 \cap L_4 = \{Q\}$

Two lines:

l : from P & Q

m : y axis

Idea of Schubert calculus:

Geometric
counting
problem



Calculation in
a ring
(cohomology ring)



Combinatorial
computation

Everything is over $\mathbb{C}$ now.

We know, $\left\{ \begin{array}{c} \text{lines in} \\ \mathbb{P}^3 \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{2-dimensional} \\ \text{subspaces of } \mathbb{C}^4 \end{array} \right\}$

Grassmannian: $Gr(k, n) := \{k\text{-dimensional subspaces of } \mathbb{C}^n\}$

Caution: Notation for Grassmannian in "Variations on a theme of Schubert Calculus" - Maria Gillespie is $Gr(n, k)$.

We consider,

$Gr(2, 4) = \{\text{planes passing through origin in } \mathbb{C}^4\}$

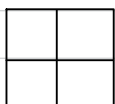
$$\dim_{\mathbb{C}} Gr(k, n) = (n-k) \cdot k$$

(This can be seen by considering each point of $Gr(k, n)$ defined as now space of $k \times n$ matrix of rank k .)

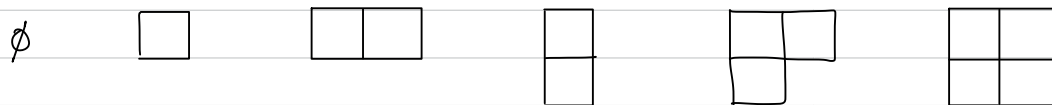
Then, $\dim_{\mathbb{C}} \text{Gr}(2,4) = 2 \cdot (4-2) = 4$

Thus, $\left\{ \begin{array}{c} \text{lines in} \\ \mathbb{P}^3 \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{points in a} \\ 4\text{-dim space} \end{array} \right\}$

We consider diagrams containing boxes that fit inside the diagram below:



We get:



These diagrams are called **Young diagrams**.

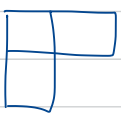
$\left\{ \begin{array}{c} \text{Young} \\ \text{diagram} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Partitions} \\ \lambda \end{array} \right\}$

Partition: Non-increasing sequence of nonnegative integers.

$\lambda = (\lambda_1, \lambda_2, \dots)$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$

$\lambda_i \longleftrightarrow$ # of boxes in i^{th} row of young diagram

- $|\lambda| := \sum \lambda_i = \text{size of } \lambda = \# \text{ of boxes}$

Ex: $(2,1) \longleftrightarrow$  $|\lambda| = 3$

$(1,0) \longleftrightarrow$  $|\lambda| = 1$
"
 (1)

Schubert varieties in $\text{Gr}(k,n)$ are indexed by partitions λ whose Young diagrams fit inside $k \times (n-k)$ rectangle.

Def: Complete flag in $\mathbb{C}^n$ is an increasing chain of subspaces $F_\bullet: F_0 \subset F_1 \subset \dots \subset F_n = \mathbb{C}^n$ where $\dim F_i = i$

Def: Schubert variety in $Gr(k, n)$:

$$\Omega_{\lambda}(F_{\bullet}) = \{V \in Gr(k, n) \mid \dim(V \cap F_{n-k+i-\lambda_i}) \geq i\}$$

Consider $\Omega_{\lambda}(F_{\bullet})$ for $\lambda = (1)$ in $Gr(2, 4)$

$\Omega_{\square}(F_{\bullet}) = \{V \in Gr(2, 4) \mid \dim(V \cap F_2) \geq 1\}$, where $F_{\bullet}$ is a fixed flag.

Let $L \subseteq \mathbb{P}^3$ be the line which corresponds to $F_2 \subseteq \mathbb{C}^4$

Then $V \in \Omega_{\square}(F_{\bullet}) \Leftrightarrow \dim(V \cap F_2) \geq 1 \Leftrightarrow L \cap V \neq \emptyset$ in $\mathbb{P}^3$,
where $L \subseteq \mathbb{P}^3$ is the line corresponding to plane V .

Thus, $\Omega_{\square}(F_{\bullet}) = \{\text{lines in } \mathbb{P}^3 \text{ meeting the fixed line } L\}$

If we consider 4 flags $F_{\bullet}^{(1)}, F_{\bullet}^{(2)}, F_{\bullet}^{(3)}, F_{\bullet}^{(4)}$ such that
 $F_2^{(i)} \subseteq \mathbb{C}^4 \iff L_i \subseteq \mathbb{P}^3$

$$\text{Then, } \bigcap_{i=1}^4 \Omega_{\square}(F_{\bullet}^{(i)}) = \left\{ \text{lines meeting 4 general lines } L_1, L_2, L_3, L_4 \text{ in } \mathbb{P}^3 \right\}$$

Thus, # lines $\leadsto$ intersection of Schubert varieties in $Gr(2, 4)$

Now we replace these varieties by algebraic objects whose multiplication will remember the intersection of varieties.

Thm: The cohomology ring $H^*(Gr(k, n))$ has a $\mathbb{Z}$ -basis given by Schubert classes

$$\sigma_{\lambda} := [\Omega_{\lambda}] \in H^{2|\lambda|}(Gr(k, n))$$

where λ fits inside $k \times (n-k)$ rectangle.

Here, multiplication is, $\sigma_{\lambda} \cdot \sigma_{\mu} := [\Omega_{\lambda} \cap \Omega_{\mu}] \in H^{2|\lambda|+2|\mu|}(Gr(k, n))$

Then, $\sigma_{(1)} = [\Omega_{\square}] \in H^2(Gr(2, 4))$ and $\sigma_{(1)}^4 \in H^8(Gr(2, 4))$

Now $H^8(Gr(2, 4))$ has basis elements σ_{λ} where $2|\lambda| = 8$ ie. $|\lambda| = 4$
and λ fits inside



$$\Rightarrow \lambda = (2, 2) \iff \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \quad \text{ie. } \sigma_{(1)}^4 = c \cdot \sigma_{(2,2)}, \text{ where } c \in \mathbb{Z}.$$

Thus, we are interested in finding the value of c .

Schur Function:

Let $\underline{x} = (x_1, x_2, \dots)$

Def: For a given partition λ ,
Schur function $s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} \underline{x}^{c(T)}$

Def: SSYT - A Semistandard Young tableau of λ is a filling of the Young diagram of λ with positive integers such that the entries are weakly increasing along the rows and strongly increasing along the columns

$\text{SSYT}(\lambda) = \{\text{all possible SSYT of } \lambda\}$

Def: $c(T) = (c_1, c_2, \dots)$ where $c_i = \#i$ in the SSYT T .

Def: $\Lambda(x_1, x_2, \dots) :=$ ring of bounded degree formal power series which are symmetric under permutation of variables.

Fact: $s_\lambda \in \Lambda(\underline{x})$, $\forall$ partitions λ and they form a vector space basis of $\Lambda(x_1, x_2, \dots)$.

Thm: $H^*(Gr(k, n)) \cong \Lambda(\underline{x}) / (s_\lambda \mid \lambda \neq k \times (n-k) \text{ rectangle})$
 $\sigma_\lambda \mapsto s_\lambda$

Thus, to find the number c such that $\sigma_{(1)}^4 = c \cdot \sigma_{(2,2)}$ we use a product rule for s_λ .

Pieri rule: For $(n) = \underbrace{\square \square \dots \square}_n$, λ a partition

$$s_{(n)} \cdot s_\lambda = \sum_{\mu \setminus \lambda \text{ is a horizontal strip of size } n} s_\mu$$

Def: Skew shape: Given two partitions λ, μ , such that $\mu \supset \lambda$ i.e. $\mu_i \geq \lambda_i, \forall i$

$\mu \setminus \lambda$: = the shape obtained by removing the Young diagram of λ from the Young diagram of μ

$E_\lambda: \mu = (4, 3, 2), \lambda = (3, 2)$



Def: A horizontal strip of size n is a skew-shape containing n boxes such that no column contains more than one box.

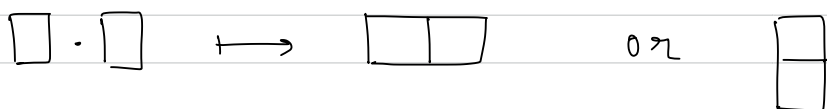
From the isomorphism above, we can apply Pieri rule for σ_λ i.e.

$$\sigma_{(n)} \cdot \sigma_\lambda = \sum \sigma_\mu$$

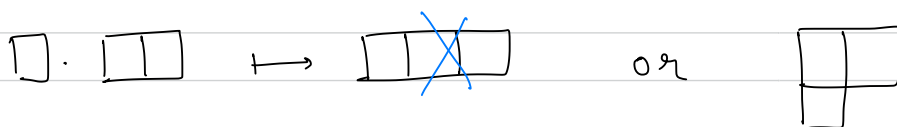
$\mu \setminus \lambda$ is a horizontal strip of size n .

To find c s.t. $\sigma_{(1)}^4 = c \cdot \sigma_{(2,2)}$.

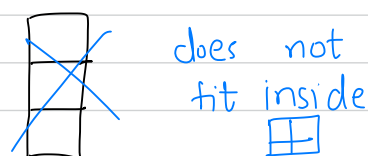
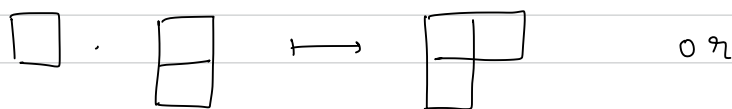
$\sigma_{(1)} \cdot \sigma_{(1)} = \sigma_{(2)} + \sigma_{(1,1)}$



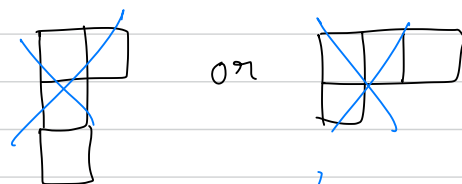
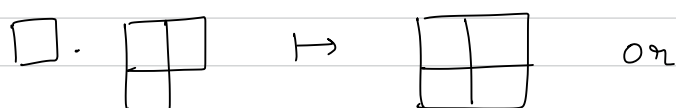
$\sigma_{(1)} \cdot \sigma_{(1)}^2 = 2 \sigma_{(2,1)}$



does not fit inside



$\sigma_{(1)} \cdot \sigma_{(1)}^3 = 2 \sigma_{(2,2)}$



does not fit inside

$\Rightarrow c=2$

- Thus, there are two lines intersecting given 4 lines in $\mathbb{P}^3$.